

ARI as a Personal Office Assistant

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Goal of the Project

The aim of this project was to employ ARI, a humanoid robot developed by PAL Robotics, as a multilingual personal assistant in an office setting. ARI was expected to follow individuals through a known environment, interpret and translate spoken language, and interact naturally through voice and motion. The project required integration of various robotic subsystems including localization using RGB-D cameras, path planning with collision avoidance, voice-based intention recognition, and natural language understanding. An essential aspect was enabling ARI to orient itself toward the user while speaking and to allow the user to direct ARI interactively. Finally, ARI was expected to search for specific locations or individuals based on prior contextual data.

Intended Learning Outcomes

David

David intended to implement mapping and localization using RGB-D input at a user-friendly level. He aimed to combine the complexities of human interaction with robotic path planning, improving both the technical and interactive aspects of navigation.

Laura

Laura planned to integrate multiple models for speech recognition, translation, and voice synthesis into a cohesive system. Additionally, she aimed to study human responses to robot behaviour from a social robotics standpoint and apply computer vision algorithms to aid in the HRI.

Joshua

Joshua wanted to investigate the nature of robot-human interaction, particularly with respect to speech. His goals included developing sound localization through microphone arrays and contributing to the creation of a detailed environmental map.

Helena

Helena’s objectives included exploring multi-modal interaction between humans and robots. She wanted to focus on the detection of user intent based on body orientation and gaze, as well as work with a robot in an unpredictable environment.

Resources Provided by PAL Robotics

The project was completed using Pal robotic’s ARI, a ”social robot to assist customers with dynamic information, run promotional campaigns and conduct research in HRI” [1]. The robot had been used by the KTH Robotics, Perception and Learning department, but all the autonomous functionalities had not been explored fully yet.

Ari has an extensive set of sensors, including 2 **RGB-D camera’s** (in the torso and head) used for obstacle avoidance and human-robot interaction, a stereo fisheye camera used for docking, and a **LIDAR** used for mapping. In addition, the robot has 2 30W Hi-Fi full-range speakers and an array of 4 high-performance digital microphone arrays, which can detect and localize sound. For interaction, the robot has a touchscreen, 2 eyes with LCD screens, 16 RGB LED’s as ”ears” and a 40 RGB LED ring.

In addition to hardware, Pal provides a large library of integrated software to aid in human-robot interaction and navigation. The library includes navigation, mapping, and obstacle avoidance. In addition, there are a lot of features related to human-robot interaction, including text-to-speech capabilities, speech-to-text, face detection and recognition, skeleton tracking, and dialogue management. When the robot is turned on, it periodically moves the arms and eyes to simulate natural behaviour.

Implementation

To manage ARI’s high-level behaviour, we implemented a modular behaviour tree system using ROS Noetic and Python. Each functionality was encapsulated in individual ROS nodes, which were coordinated by the behaviour tree. This architecture supported flexible task composition and reliable execution.

The key modules developed were:

- Face recognition for identifying and tracking individuals.
- Speech-to-text (STT) for interpreting user commands in various languages.
- Text-to-speech (TTS) for enabling ARI to respond vocally in various languages.
- A system that allows ARI to track and walk behind a person.

- Orientation control was implemented to ensure that ARI could face the speaker during conversations and position herself beside them, aligned in the same direction. This allowed ARI to function naturally as a translator—standing next to the speaker and facing the same audience—making the interaction more intuitive and human-like.
- Navigation capabilities to move to predefined room locations.
- Speaker localization combining voice and body cues.
- Looking at a person when in a conversation.
- Two-way translation

A key innovation was the integration of a Large Language Model (LLM) for natural language understanding. This allowed users to interact with ARI conversationally, without using fixed command phrases. The LLM parsed the intent behind the speech and routed it to the corresponding behaviour tree action, improving usability in various office interactions. Each intent is wrapped in a reusable structure that tracks execution success and failure, with a fail-safe mechanism halting the action after 10 failed attempts. The tree is ticked left to right every second, giving higher priority to left-most actions (e.g., “stop”), which can interrupt lower-priority tasks (e.g., “go to”). This design allows flexible, reusable, and hierarchical control of ARI’s actions with clear fallback behaviours.

Results and Reflections

The project successfully demonstrated the concept of ARI as a mobile office assistant and translator. The robot could reliably follow people, interpret conversational instructions, and maintain orientation toward the speaker. Language translation was achieved using online services, and most interactions were handled naturally. She can successfully save people’s location and go to the room they were last seen in when asked to find that specific person. When asked to be guided to a specific room set on the office map, she will reach it.

Multi-body tracking was achieved as a separate node but not implemented in the final demo since most of the modules had already been made and tested in single body detection. However, we believe this module to be very interesting for future projects with ARI.

Challenges and Lessons Learned

Several issues and challenges arose during development.

Face Recognition

Blurry images caused by motion sometimes led to the creation of duplicate “new people” in the face recognition dataset, increasing the computational load during comparisons. To solve this, we stored multiple face encodings per person, improving recognition robustness.

This significantly reduced false positives, although not entirely. Additionally, in rare cases, a person’s location could not be accurately estimated if they were too far from the camera, preventing ARI from identifying their presence in any room.

This challenge provided valuable insight into the practical limitations of face detection in real-time systems, especially when dealing with video-based recognition under non-ideal conditions.

Microphone Positioning

The front-facing microphones made it hard to detect speakers behind ARI, especially in echo-prone rooms. We solved this by using the microphones only to detect speech and estimate a rough direction. ARI then turns and stops once someone is seen by the head camera. This showed the value of combining audio and visual input.

Language Support

Since only English was supported and PAL’s multilingual API was unavailable, we integrated third-party TTS and STT modules to enable ARI to understand and speak multiple languages—essential for our translator goal in RPL’s international setting. This challenge gave us valuable insight into the world of speech systems, as we explored and tested different modules and learned a lot from the process.

Obstacle Navigation

Navigating narrow office doors required adjusting ARI’s obstacle radius to enable tight turns. We also mapped “door zones” to trigger this adjustment, lower the radius when passing through and restoring it back to its original value when the doorway has ended. Most rooms needed ramps, and the MOCAP room’s step was impassable even with one. This taught us the importance of real-world testing and fine-tuning navigation parameters to fit the environment, and how difficult navigating unknown environments really is.

Mapping

The mapping functionality discovered several complications with the use of the LiDAR and the RGBd cameras to build an accurate map of the environment. Glass was one of the most difficult elements to deal with, as it is not detected by LiDAR and the cameras so easily. The algorithms already implemented by PAL highlighted some issues with permanence of obstacles even after said obstacles have been removed.

Planning

Due to time constraints a compromise on the generation of navigation points had to be made, and as such there was not a cross checking against the map to determine whether the

generated points were collision free or not, this led to situations where ARI would freeze as it was instructed to go into a zone where the inflation layer or an actual obstacle was present.

Localization

The pre-existing algorithm given by PAL is based on a Particle filter and it performs well enough to generate accurate localization parting from a evenly spread out distribution, but as the map grew larger the results tend to became harder to achieve as the particle count was fixed. This type of algorithm also presented one of the most common problems when it comes to determining the robot’s position, this issue was most present in long hallways as the lack of markers would create larger uncertainty of the robot’s position as it moved through the environment.

Hardware malfunctions: Eyes and Screen

Since the beginning, both the eyes and the screen were non-functional. After contacting PAL Robotics, the issue with the eyes was successfully resolved. However, the Chrome screen remains inoperative, which has occasionally limited our ability to display useful information to the speaker.

Unfiltered Voice Input

Initially, ARI transcribed all speech in the environment, which overwhelmed the LLM with irrelevant inputs. We solved this by adding a wake-word trigger (“robot”) that activates listening for 10 seconds, improving stability and accuracy. “Stop” remains an interrupt command. This gave us valuable insight into designing robust voice interfaces in dynamic, noisy environments.

Network and Computing constraints

Partly due to the projects architecture design, some of the components were too heavy in the consumption of computing resources to be handled by ARI alone or even a single machine. So the workload had to be divided among ARI and two other laptops. This limited somewhat the responsiveness of ARI to certain actions as it had to wait for answers to come through the network, this also imposed a constraint in which the architecture of the system had to be though in such a way that would minimize the use of bandwidth in the network. This readuction in use of bandwidth became a priority when changing the robot to work in the KTH-IoT network as the throughput of the network would fluctuate every day.

Conclusions

This project emphasized the importance of multi-modal integration in human-robot interaction. We found that even state-of-the-art components such as SLAM or speech recognition require meticulous tuning and coordination when deployed in the real world. Behaviour trees

proved effective for managing complex tasks and modularizing robot functionality.

Perhaps most critically, we discovered that the use of LLMs significantly improved the user experience by allowing natural, conversational interaction with the robot. Beyond technical challenges, we also learned about the nuances of user perception and social dynamics in human-robot interaction, setting the foundation for future research into socially aware autonomous systems.

One of the most valuable takeaways from this project has been applying our knowledge to a real robot, where we encountered numerous unexpected challenges. Many components didn't work out of the box or required meticulous adjustments, highlighting how difficult it can be to operate in unpredictable, real-world environments.

References

- [1] ARI — Social and Collaborative Robot for Research & Entertainment — pal-robotics.com. <https://pal-robotics.com/robot/ari/>. [Accessed 28-05-2025].